Q7. Backoff Model (based on "Activity: <s> I like cats … You like dogs" slide)

Training corpus:

<s> I like cats </s>
<s> I like dogs </s>
<s> You like cats </s>

Counts:

- I like = 2
- You like = 1
- like cats = 2
- like dogs = 1
- cats </s> = 2
- dogs </s> = 1

Tasks:

1. Compute P(cats|I,like).

   P(cats|I,like)=count(I,like,cats)/count(I,like)
- Count(I like cats) = 1 (from sentence `<s> I like cats </s>`)
- Count(I like) = 2
- P(cats|I,like) = 1/2 = 0.5

2. Compute P(dogs|You,like) using trigram → bigram backoff.

   First check trigram:
   P(dogs|You,like) = count(You,like,dogs)/count(You,like)

   Count(You like dogs) = 0 (never appears)

   Count(You like) = 1
   So trigram = 0/1 =0.

   Backoff to bigram.
   Bigram probability: P(dogs|like) = count(like,dogs)/count(like)
- Count(like dogs) = 1
- Total after "like" = count(like cats) + count(like dogs) = 2 + 1 = 3

   P(dogs|like) = 1/3 ≈0.333

3. Explain why backoff is necessary in this example.

   Because the training corpus is tiny, many valid trigrams never appear (e.g., "You like dogs").
- MLE assigns probability = 0 to unseen trigrams.
- This makes entire sentence probabilities = 0 (zero-probability problem).
- **Backoff** solves this by "falling back" to a lower-order model (bigram, unigram) that can still provide a non-zero estimate.